

COMPUTER NETWORK....

ASSIGNMENT-01

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DEPT : III - CCE.....

SUB CODE : 23CST501....

TITLE : TRANSPORT LAYER.....

DATE : 13/07/2025....

Justify the necessity of the transport layer in network communication.

Transport layer:

Transport layer is the 4th layer in the OSI model and is responsible for delivering messages between host systems.

The transport layer, one of the core layers in the OSI (Open Systems Interconnection) model, plays a pivotal role in enabling data communication between devices would be unstructured and prone to errors. It acts as a bridge between the application and network layers, ensuring the data is delivered correctly, completely and in order.

Role of the Transport layer.

The primary function of the transport layer is to provide end to end communication between devices, unlike the network layer, which deals with routing and forwarding of data packets across network, the transport layer focuses on establishing a connection

detect and correct error. while the physical data link layer also contribute to error detection, they are limited to the local network. the transport layer, however, provides end to end error control to add checksum to segments and ensuring that the receiver acknowledges the successful reception of data.

Flow control.

Data can be sent faster than the receiver can process it. To avoid overwhelming the receiving device, the transport layer uses flow control mechanisms. TCP for instance, uses a sliding window protocol to control how much data can be sent before receiving an acknowledgement. This ensures that the sender does not flood the receiver, maintaining a balance in communication.

Congestion Control.

Network congestion can degrade the performance of applications significantly. The transport layer, particularly in TCP

implements congestion control algorithm such as slow start, congestion avoidance & fast recovery. These algorithms help prevent congestion by adjusting the data transmission rate based on network condition.

Real World Applications of the Transport Layer.

To further understand the necessity of a transport layer, it's helpful to examine how its function in real world application.

eg:

When a user stream a movie on platform like Netflix or Youtube, the transport layer ensures that video data is delivered at a consistent data rate. Although service may use UDP for lower latency.

This protocol not only encrypt data but also validate the identity of communicating parties - ensuring that data reaches the correct recipient safely.

between the source and destination hosts. It enables process to process communication, meaning it connect specific applications on one machine to specific application on another.

Connection-oriented and Connectionless services.

The transport layer support both connection-oriented and connectionless (TCP) (UDP) services. TCP ensures that data is delivered reliably and in sequence. It uses mechanisms like acknowledgement, retransmission, and flow control to provide error free delivery. UDP (User datagram protocol) on the other hand, is faster but does not guarantee reliability, making it ideal for application like video streaming or online gaming and speed is more critical than the reliability.

Error detection & correction

One of the major justifications for the transport layer is its ability to

implements congestion control algorithms such as slow start, congestion avoidance and fast recovery. These algorithm help prevent congestion by adjusting the data transmission rate based on network conditions, ensuring smooth communication without overwhelming the network.

Segmentation & Reassembly.

Large messages must be broken into smaller packets for transmission across network.

The transport layer is responsible for segmenting data into smaller units and assembling them at Receiving end. This allows the network to handle data efficiently, especially when transferring files or streaming large media content.

Port number and multiplexing :-

The transport layer uses port numbers to distinguish between different applications running on the same device. This is essential in modern computing environments where multiple applications communicate simultaneously, multiplexing and